

significant advances in modern automation is the shift towards edge AI. Instead of sending all data to the cloud for processing, which can be slow, costly, and unreliable, machines now process information locally at the source.

A robot that detects a defect and immediately adjusts its actions without waiting for external processing demonstrates edge AI in action. Decisions are made in real time, which is critical for high-speed production lines or when handling delicate products where even small delays can cause errors.

Edge AI also improves reliability. Machines continue operating even without an internet connection. Data security is strengthened because information remains inside the factory, and costs are reduced through lower reliance on cloud infrastructure.

- **Visual inspection systems.** The built-in cameras can detect surface defects, misalignment, or missing parts, which can be corrected immediately without stopping the line.

- **Predictive maintenance.** You can track vibration, sound, and temperature with edge AI. It also helps fix the problem before a breakdown.

- **Adaptive robotics.** Live sensors help robots adjust their grip strength, speed, and movement, thereby increasing both safety and efficiency.

- **Collaborative robots (cobots).** Edge AI enables cobots to respond quickly to human movements. This makes working with robots safer than ever before.

Consider Epson's RC+ platform, which combines motion control, vision, and data analysis into a single system. This allows robots to adapt to slight changes in any component or environment. Or consider Siemens' SIMATIC with Industrial Edge, which gives the processing power of edge computing to PLCs, so factories can predict maintenance requirements and identify problems before they cause downtime.

What makes an engineer 'industry-ready' for robotics today

Modern robotics roles demand more than basic programming skills. An industry-ready robotics engineer is expected to understand:

- Robot programming and commissioning (SCARA and articulated robots)
- Vision system calibration and AI-based inspection logic
- Edge AI concepts for adaptive automation and predictive maintenance
- Safety standards (collaborative operation, risk assessment)
- System integration with PLCs, conveyors, and factory networks
- Troubleshooting in live production environments

Hands-on exposure to real industrial systems, rather than simulations alone, differentiates employable graduates from those with only theoretical training. This is why production-grade labs and internships at solution centres are becoming central to engineering education strategies.

Rockwell Automation integrates its Logix control lines with FactoryTalk Analytics, enabling machines to detect anomalies and self-correct. Omron's Sysmac platform integrates robotics, vision, and AI, enabling robots to do more than follow a path; they can adjust paths, compensate for differences, and even work alongside humans.

Edge AI is turning robots into practical problem-solvers on the factory floor. A packaging robot can detect a misaligned box and adjust its grip instantly. A quality inspection system can identify defects and trigger corrective action upstream. As autonomy increases, robots will operate more independently, maintain performance under changing conditions, and require less intervention, with edge intelligence serving as the enabling foundation.

Vision-led automation becomes the default

Modern robots increasingly appear to 'see' the world around them. Machine vision has evolved from a desirable but optional feature into a foundational capability of modern robotics. In sectors such as electronics, packaging, and inspection-intensive manufacturing, robots without vision are comparable to cars without headlights: they may move, but they cannot reliably navigate real-world conditions. The real value of vision emerges when it is combined with

AI. Platforms such as Keyence's CV-X and XG-X, Omron's FH and FZ, Cognex In-Sight cameras, and integrated vision solutions from robot manufacturers are now paired with SCARA and six-axis systems to create intelligent, flexible automation cells.

Vision-guided robotics enables tasks that were previously difficult or impractical to automate. In random bin picking, robots identify and grasp parts without a fixed orientation. In inline quality inspection, cosmetic defects are detected, and products are flagged before shipment. Adaptive assembly, where adjustments are required based on part variation, is also increasingly feasible. Cameras and sensors allow robots to interpret their surroundings, adapt to variability, work more efficiently, and reduce errors.

Where robotics is expanding next

Robotics adoption is no longer confined to a single industry or use case. Its footprint is widening across manufacturing and logistics, with each sector applying automation in distinct, practical ways.

Electronics and semiconductor manufacturing. Robots are widely used in electronics manufacturing in India. Thanks to the Production-Linked Incentive schemes initiated by the government of India. Currently, robots are used to place components on circuit boards, insert connectors, and operate in clean